FORM 2

PATENTS ACT, 1970

(39 of 1970)

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION

"A MINIATURIZED ENVIRONMENTAL MONITORING SATELLITE-CAN SAT"

2. APPLICANTS

a)	**Name**	**:**	**ANURAG UNIVERSITY**
b)	**Nationality**	**:**	**INDIAN**
c)	**Address**	**:**	**Venkatapur, Ghatkesar, Medchal- Malkajgiri District, Hyderabad, Telangana-500088.**

3. PREAMBLE TO THE DESCRIPTION

COMPLETE

The following specification particularly describes the invention and the manner in which it is to be performed.

4. DESCRIPTION

Technical Field of the Invention

The present invention relates to aerospace engineering and environmental monitoring, specifically to a miniaturized satellite system (CanSat) designed for real-time atmospheric and aquatic data collection using advanced sensors, communication modules, and GPS tracking. It provides a cost-effective solution for applications in scientific research, disaster management, pollution tracking, and education.

Background of the Invention

Environmental monitoring is essential for understanding climate change, pollution levels, and atmospheric variations. Traditional satellite-based monitoring systems provide valuable data but are often costly, require extensive infrastructure, and are not accessible for small-scale research and educational purposes. Miniaturized satellite systems like CanSats have emerged as a cost-effective alternative, enabling real-time environmental data collection. However, existing CanSat models face several limitations, including restricted onboard memory, communication range constraints, limited power efficiency, and inaccuracies in GPS tracking.

Additionally, many current CanSat designs are primarily focused on atmospheric monitoring, limiting their applicability to aquatic environments. Monitoring water quality parameters, such as temperature, dissolved solids, and pollution levels, is crucial for environmental conservation and disaster response. However, existing solutions lack effective waterproofing, buoyancy control, and long-duration operational capability in water bodies.

The present invention overcomes these challenges by integrating an advanced multi-sensor suite for atmospheric and aquatic monitoring, long-range wireless communication, GPS tracking with improved accuracy, and an optimized power

management system. It introduces custom-designed Yagi and dipole antennas for enhanced communication range, a hybrid buoyancy mechanism for stable flotation, and an improved recovery system using LED indicators and a buzzer.

By addressing these challenges, the invention provides a cost-effective, scalable, and reliable solution for scientific research, disaster management, pollution tracking, and aerospace education. It enhances real-time data collection efficiency, ensuring broader accessibility for environmental studies and practical training in satellite technology.

Brief Summary of the Invention

The present invention relates to a low-cost, miniaturized CanSat system designed for real-time environmental monitoring in both atmospheric and aquatic conditions. The Can Sat is equipped with a multi-sensor suite, including temperature, humidity, pressure, air quality, and water contamination sensors, ensuring accurate data collection.

OBJECTS OF THE INVENTION:

The primary objective of the invention is to develop a low-cost, miniaturized satellite system (CanSat) capable of real-time environmental monitoring in both atmospheric and aquatic conditions with high accuracy and reliability.

Another objective is to enhance data transmission and retrieval efficiency by integrating custom-designed Yagi and dipole antennas, long-range wireless communication modules, and GPS tracking with LED indicators and a buzzer for easy recovery.

Another objective is to improve sensor accuracy and data logging capabilities by utilizing an advanced multi-sensor suite that measures atmospheric parameters such as

temperature, humidity, air quality, and pressure, along with aquatic parameters like water temperature and total dissolved solids (TDS).

Another objective is to extend operational endurance and optimize power consumption by incorporating a high-capacity Li-Po battery with an intelligent charging system, ensuring long-duration performance in remote environmental conditions.

Another objective is to develop a scalable, cost-effective model that can be used for scientific research, disaster management, pollution tracking, aerospace education, and defense reconnaissance applications.

Another objective is to introduce a hybrid buoyancy mechanism to ensure stable flotation and continuous operation in aquatic environments, making the CanSat suitable for both air and water-based environmental monitoring.

SUMMARY OF THE INVENTION:

The present invention relates to a low-cost, miniaturized CanSat system designed for real-time environmental monitoring in both atmospheric and aquatic conditions. The CanSat is equipped with a multi-sensor suite, including temperature, humidity, pressure, air quality, and water contamination sensors, ensuring accurate data collection.

The invention incorporates custom-designed Yagi and dipole antennas for extended communication range, along with GPS tracking, LED indicators, and a buzzer for easy retrieval. A hybrid buoyancy mechanism allows stable operation in water, enhancing its dual-environment functionality.

To ensure prolonged operation, the system includes a high-capacity Li-Po battery with an intelligent charging controller, optimizing power consumption. The invention is

designed to be cost-effective, scalable, and suitable for research, education, disaster monitoring, pollution tracking, and defense reconnaissance.

By integrating real-time data logging, long-range communication, and advanced recovery mechanisms, the CanSat offers a versatile and reliable solution for environmental monitoring and aerospace training applications.

APPLICATIONS OF THE INVENTION:

The invention finds significant application in atmospheric and aquatic environmental research, enabling real-time monitoring of air and water quality parameters. By providing precise measurements of temperature, humidity, air pollutants, and water contamination levels, it supports scientific studies focused on environmental sustainability and conservation.

In educational training in aerospace and robotics, the CanSat serves as an effective tool for students and researchers to gain hands-on experience in satellite technology, sensor integration, and real-time data transmission, fostering innovation in space and environmental sciences.

For disaster monitoring and pollution tracking, the system aids in assessing air and water pollution levels, detecting hazardous gases, and monitoring environmental changes during natural calamities such as wildfires, floods, and industrial accidents, assisting authorities in disaster management.

In remote environmental sensing for climate studies, the invention contributes valuable meteorological and hydrological data for analyzing climate change patterns, helping researchers understand long-term environmental variations and their impact on ecosystems.

The CanSat also has potential applications in defense and military reconnaissance, where it can be deployed for surveillance, data collection, and environmental assessment in strategic locations, providing crucial real-time intelligence.

In industrial and agricultural applications, the system assists in monitoring environmental conditions for pollution control in industries and ensures water quality management in agriculture, thereby improving crop yield and maintaining ecological balance.

For smart city and urban monitoring, the invention contributes to smart city initiatives by tracking air pollution levels, weather conditions, and environmental hazards, supporting the development of sustainable and healthier urban environments.

Overall, the invention offers a cost-effective, scalable, and versatile solution for a wide range of environmental monitoring applications, benefiting scientific research, public safety, and sustainable development.

ADVANTAGES OF THE INVENTION:

The invention provides a cost-effective alternative to traditional satellite-based environmental monitoring systems. By utilizing miniaturized components and efficient design, it significantly reduces development and operational costs while maintaining high functionality.

The dual-environment functionality allows the CanSat to operate in both atmospheric and aquatic conditions, making it a versatile tool for comprehensive environmental monitoring across different terrains and ecosystems.

The system features extended operational endurance with an efficient power management system, including a high-capacity Li-Po battery and intelligent charging controller, ensuring long-duration operation without frequent recharging.

An enhanced recovery mechanism, incorporating GPS tracking, LED indicators, and a buzzer, ensures easy retrieval after deployment, reducing the chances of loss and enabling repeated use.

The custom-designed Yagi and dipole antennas provide an extended communication range, improving real-time data transmission over long distances, making the system suitable for remote monitoring applications.

The invention employs automated real-time and long-term data logging, enabling continuous environmental data collection for scientific research, climate studies, and industrial monitoring without the need for manual intervention.

With its scalable and modular design, the CanSat can be easily adapted for research, educational training, disaster management, pollution tracking, and defense applications, offering a highly flexible and practical solution for various industries and domains.

Brief Description of the Drawings

The invention will be further understood from the following detailed description of a preferred embodiment taken in conjunction with an appended drawing, in which:

FIG. 1a-1b illustrates the miniaturized environmental monitoring satellite (Can Sat 100) designed for atmospheric and aquatic data collection, in accordance with an exemplary embodiment of the present invention;

Fig. 2 illustrates the system's operational flow, in accordance with an exemplary embodiment of the present invention;

Fig. 3 provides a detailed step-by-step process of the mission, in accordance with an exemplary embodiment of the present invention;

Detailed Description of the Invention

It is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The present disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. In addition, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

The use of “including”, “comprising” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. The terms “a” and “an” herein do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced items. Further, the use of terms “first”, “second”, and “third”, and the like, herein do not denote any order, quantity, or importance, but rather are used to distinguish one element from another.

According to an exemplary embodiment of the present invention, a Low-Cost Innovative CanSat for Environmental Monitoring is provided, designed for real-time atmospheric and aquatic data collection. The system integrates multiple sensors, communication modules, and recovery mechanisms, ensuring efficient environmental analysis. The invention offers a compact, cost-effective, and scalable solution for scientific research, disaster management, and educational applications.

According to an exemplary embodiment of the present invention, the ATmega328P Pro Mini serves as the central processing unit (CPU) of the CanSat, executing programmed instructions and managing sensor data collection, processing, and transmission. It is based on Harvard architecture with RISC (Reduced Instruction Set Computing), ensuring power efficiency and optimized computing performance.

In accordance with an exemplary embodiment of the present invention, the BME680 sensor is employed to measure temperature, humidity, pressure, and volatile organic compounds (VOCs). It operates through barometric pressure detection and gas sensing technology, enabling precise environmental assessments.

According to an exemplary embodiment of the present invention, the MICS5524 sensor is integrated to detect harmful gases such as carbon monoxide (CO), nitrogen dioxide (NO_2), and ammonia (NH_3). The sensor functions based on a chemical reaction mechanism, where resistance variations indicate pollutant levels, providing real-time air quality monitoring.

In accordance with an exemplary embodiment of the present invention, the ATGM336H GPS module provides real-time geolocation data using satellite triangulation techniques. It assists in tracking the CanSat's position, ensuring accurate recovery post-deployment.

In accordance with an exemplary embodiment of the present invention, wherein the Total Dissolved Solids (TDS) sensor is utilized to measure water purity and contamination levels. It operates by analyzing electrical conductivity in water, determining impurity concentrations for aquatic environmental monitoring.

In accordance with an exemplary embodiment of the present invention, the DS18B20 digital temperature sensor is integrated for aquatic thermal monitoring. The sensor

functions using one-wire digital communication to provide high-precision temperature readings, aiding in climate and oceanographic studies.

In accordance with an exemplary embodiment of the present invention, the MPU6050 combines a gyroscope and accelerometer using MEMS (Micro-Electro-Mechanical Systems) technology to monitor motion, stability, and orientation of the CanSat during deployment and descent.

In accordance with an exemplary embodiment of the present invention, the HC-12 wireless transceiver operates at 433 MHz frequency band, allowing long-range serial data communication up to 3 km. This ensures seamless data transmission to the ground station.

In accordance with an exemplary embodiment of the present invention, a 5000mAh Lithium-Polymer (Li-Po) battery is used as the primary power source. It efficiently converts chemical energy to electrical energy, providing extended operational life to the Can Sat.

In accordance with an exemplary embodiment of the present invention, the TP4050 battery charge controller regulates charge cycles, preventing overcharging and deep discharge, ensuring battery longevity and operational stability.

In accordance with an exemplary embodiment of the present invention, capacitors and resistors function as voltage stabilization and energy dissipation elements in the circuit. They help maintain consistent power delivery, reducing voltage fluctuations and ensuring smooth system performance.

In accordance with an exemplary embodiment of the present invention, LED indicators utilize electron-hole recombination technology to emit light, providing visual feedback for system health, power status, and data transmission activity.

The invention further includes optimized waterproofing techniques, hybrid buoyancy mechanisms, and an advanced master-slave microcontroller framework, ensuring superior performance in environmental monitoring, research, and real-time data collection applications.

Referring to Fig's, FIG. 1 illustrates a miniaturized environmental monitoring satellite (CanSat 100) designed for atmospheric and aquatic data collection. The CanSat (100) comprises a primary control unit (102), which includes an ATmega328P Pro Mini microcontroller responsible for executing pre-programmed instructions and managing sensor data processing. Integrated within the system is a multi-sensor suite (104) that enables comprehensive environmental monitoring. The suite includes a BME680 sensor (106) for detecting temperature, humidity, pressure, and volatile organic compounds (VOCs), a MICS5524 air quality sensor (108) for analyzing gaseous pollutants such as CO, NO_2, and NH_3, and an ATGM336H GPS module (110) for real-time geolocation tracking. Additionally, for aquatic data collection, a TDS sensor (112) assesses total dissolved solids, while a DS18B20 sensor (114) measures water temperature. An MPU6050 sensor (116) is incorporated to monitor motion and stability through accelerometer and gyroscope functions.

The CanSat (100) further comprises a communication module (118) that facilitates real-time data transmission using an HC-12 wireless transceiver operating at 433 MHz. To enhance signal strength and range, custom-designed Yagi and dipole antennas are integrated into the system. A power management system (120) ensures prolonged operation using a 5000mAh Li-Po battery, regulated by a TP4050 charge controller to prevent overcharging and optimize power consumption. Additionally, the recovery

mechanism (122) is designed for easy retrieval post-mission, incorporating GPS tracking, LED indicators, and a buzzer for enhanced visibility and location identification. The CanSat (100) also features a hybrid buoyancy mechanism to maintain stability during aquatic deployment, ensuring efficient data collection.

The overall system architecture is optimized for real-time data acquisition, wireless transmission, extended operational endurance, and enhanced retrieval capabilities, making the CanSat (100) suitable for diverse environmental monitoring applications.

Fig. 2 illustrates the system's operational flow, starting from initialization to data transmission and recovery. The process begins with system initialization, where all necessary components are powered on. The sensors are activated to start collecting data, which is then processed by a slave microcontroller and forwarded to the master microcontroller. The master microcontroller transmits the collected data via HC-12 communication to the ground station. Simultaneously, the GPS system updates the position continuously to ensure accurate tracking. Finally, upon landing, the recovery mechanisms activate, ensuring safe retrieval of the system.

Fig. 3 provides a detailed step-by-step process of the mission, starting from pre-launch checks to post-mission analysis. The process begins with pre-launch checks, which include verifying the battery, calibrating sensors, and testing communication. Once everything is confirmed, the launch sequence initiates, followed by the deployment of the CanSat. Upon deployment, a parachute is deployed to control descent. During descent, sensor data acquisition begins, and data is transmitted to the ground station. The GPS tracking system continuously updates the position to assist in retrieval. Upon landing, recovery mechanisms activate, ensuring safe retrieval of the CanSat. Finally, post-mission data retrieval and analysis are conducted to evaluate the collected data and performance metrics.

5. CLAIMS

We Claim

1. A miniaturized environmental monitoring satellite (Can Sat) (100), comprising:

   a primary control unit (102) utilizing an ATmega328P Pro Mini microcontroller for data processing and execution of pre-programmed instructions;

   a multi-sensor suite (104), including:

      a BME680 sensor (106) for measuring temperature, humidity, pressure, and volatile organic compounds (VOCs);

      a MICS5524 air quality sensor (108) for detecting pollutants such as CO, NO_2, and NH_3;

      an ATGM336H GPS module (110) for real-time geolocation and tracking;

      a TDS sensor (112) for assessing total dissolved solids in aquatic environments;

      A DS18B20 sensor (114) for precise water temperature measurement;

      an MPU6050 sensor (116) incorporating an accelerometer and gyroscope for motion and stability analysis;

   a communication module (118), including an HC-12 wireless transceiver operating at a 433 MHz frequency band for long-range data transmission;

   a power management system (120), comprising a 5000mAh Li-Po battery and a TP4050 battery charge controller to regulate power consumption and prevent overcharging;

   a recovery mechanism (122), featuring GPS tracking, LED indicators, and a buzzer to facilitate easy retrieval of the device post-mission.

2. The Can Sat (100) as claimed in claim 1, wherein the device is integrated with custom-designed Yagi and dipole antennas to extend the communication range for real-time data transmission.

3. The Can Sat (100) as claimed in claim 1, wherein the system employs an optimized master-slave microcontroller framework to efficiently manage sensor data acquisition, processing, and wireless transmission.

4. The Can Sat (100) as claimed in claim 1, wherein the environmental monitoring system is configured for both atmospheric and aquatic data collection, with waterproofing techniques to prevent water ingress.

5. The Can Sat (100) as claimed in claim 1, wherein the device incorporates a hybrid buoyancy mechanism to ensure stable flotation during aquatic monitoring operations.

6. The Can Sat (100) as claimed in claim 1, wherein the data acquisition system is designed for real-time and long-term environmental data logging, enabling extended-duration monitoring.

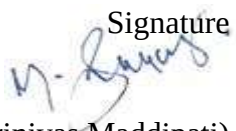
7. The Can Sat (100) as claimed in claim 1, wherein the power management system (120) is optimized to ensure extended operational endurance with minimal energy consumption.

8. The Can Sat (100) as claimed in claim 1, wherein the retrieval system is enhanced with LED indicators and a buzzer for improved visibility and tracking, even in low-light conditions.

6. DATE AND SIGNATURE

Dated this 18th March, 2025

Signature



(Mr. Srinivas Maddipati)

IN/PA 3124

Agent for Applicant

TITLE: "A MINIATURIZED ENVIRONMENTAL MONITORING SATELLITE-CAN SAT"

7. ABSTRACT

The present invention relates to a Low-Cost Innovative CanSat for Environmental Monitoring (100), designed for real-time atmospheric and aquatic data collection. The system comprises a primary control unit (102) utilizing an ATmega328P Pro Mini microcontroller for sensor data processing and command execution. A multi-sensor suite (104) is integrated, including a BME680 sensor (106) for temperature, humidity, pressure, and VOC detection, a MICS5524 sensor (108) for air quality monitoring, an ATGM336H GPS module (110) for real-time geolocation, a TDS sensor (112) for water quality analysis, a DS18B20 sensor (114) for aquatic temperature measurement, and an MPU6050 sensor (116) for motion stability assessment. A communication module (118) employing an HC-12 wireless transceiver ensures long-range data transmission, while a power management system (120) with a 5000mAh Li-Po battery and TP4050 charge controller enables extended operation. The retrieval mechanism (122) is equipped with GPS tracking, LED indicators, and a buzzer for easy location post-deployment. The invention features custom-designed Yagi and dipole antennas for enhanced signal transmission, hybrid buoyancy for stable flotation, and waterproofing for aquatic monitoring.

The figure associated with the abstract is Fig.1